Claire (Shao-yu) Lin

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Education

Carnegie Mellon University - MS in Electrical and Computer Engineering - AI/ML Systems *Pittsburgh, PA*
Courses: Deep Learning, Pattern Recognition, Distributed Systems, Computer Networks, Cloud Infrastructures *Aug 2022 - Dec 2023*

San Jose State University - BS in Computer Science, Mathematics Minor *San Jose, CA*
Courses: Machine Learning, Numerical Analysis, Engineering Statistics, Computer Architecture, Operating Systems *Aug 2016 - Aug 2021*

Skills

Programming Python, Java, C/C++, JavaScript, Scala, Matlab, R, SQL, Verilog
Tools/Frameworks PyTorch, Tensorflow, Amazon Web Services, Docker, Kubernetes, React.js, Node.js, FastAPI, Git, Linux, Jenkins
Technical Skills Anomaly detection, Medical Imaging, Natural Language Processing, Model Deployment

Work Experience

Amazon *Seattle, WA*
SOFTWARE DEVELOPMENT ENGINEER *Oct 2024 - Present*
- Designed and implemented agent AI workflow features, building REST APIs and event-driven backend services using AWS technologies
- Refined feature requirements with product managers, focusing on scalable infrastructure and improved user experience

Veytel *Pittsburgh, PA*
MACHINE LEARNING ENGINEER (CONSULTANT) | IMAGING TEAM *Mar 2024 – Oct 2024*
- Built end-to-end pipelines for image segmentation & object detection models for lesion detection, from pre-processing, training to evaluation
- Leveraged transfer learning to redesign detection models with Detectron2 and Numpy, resulting in a 34.5% increase in model accuracy

Adobe *San Jose, CA*
SOFTWARE ENGINEER INTERN | ADOBE SIGN *May 2023 – Aug 2023*
- Designed and built new eSign mobile application for offline package delivery, targeting user base of 12% of US population with spotty Internet
- The mobile app is a progressive web application built using Adobe's React design system and supported by Sign microservices backend

Clario *San Mateo, CA (remote)*
SOFTWARE ENGINEER | MEDICAL IMAGING TEAM *Aug 2020 – May 2022*
- Productionized CNN-based medical segmentation models, automating radiologists' workflows and reducing labor time by 5,000+ hours
- Collaborated with scientists, conducted experiments, and implemented over 15 image analysis pipelines using Python and C++
- Implemented company's first brain white matter parcellation algorithm via image processing endpoints
- Integrated data pipelines to cloud-based computation resources utilizing Docker, AWS Batch, and AWS EBS

Research Experience

Carnegie Mellon University *Pittsburgh, PA*
RESEARCH INTERN | ADVISOR: DR. MING XU *Dec 2023 – Present*
- Evaluated 5 text-guided diffusion models for prompting strategies and researched variational auto-encoders and depth estimation models
- Utilized PIL, Scipy, and Numpy in dataset processing. Processed 100K+ images in 20+ datasets

Projects

Wild Fire Detection *SJSU, CS 185C Machine Learning | Sep 2019 – Dec 2019*
- Developed classification model to forecast wildfire occurrence given 26 daily weather measurements, achieving accuracy of 0.89
- Trained SVM and neural network models. Processed 1.8 million SQLite entries in 5+ datasets [link]

DigiDraper | 3D Vision *Sep 2022 – Dec 2022*
- Developed virtual clothing try-on system to reconstruct 3D garment meshes (SMPL) from a single 2D body image
- Utilized self-supervised learning techniques for garment draping, adaptable to pose and material [link]

Distributed file system *CMU, 14736 Distributed Systems | April 2023*
- Built distributed file system in Java with multiple storage and naming servers, implementing read-write locks and data replication

Publications

Metric from Human: Zero-shot Monocular Metric Depth Estimation via Test-time Adaptation *NeurIPS*
Y ZHOU, H BIAN*, K CHEN*, L QU*, P JI, S LIN*, W YU, H LI, H CHEN, J SHEN, B RAJ, M XU *(2024)*
- Conducted experiments to optimize prompting strategies for stable diffusion to produce human paintings measurable by downstream Human Mesh Recovery (HMR). Achieved a 41.4% accuracy improvement on NYU and KITTI datasets compared to other zero-shot models [link]

Automated MRI Face-Removal Pipeline to Anonymize Patient Scans for Clinical Trials *CTAD 2021*
L KIDZINSKI, T CAJGFINGER, K THOMAS, L BRACOUD, S LIN, C CONKLIN ET AL. *(2021). S83-S84.*
- Performed large-scale clinical validation with 200+ subjects and verified the face-removal algorithm's 99% effectiveness [link]